Parent's Signature:

CANDIDATE'S NAME: ______________________________ **CTG:** _____________

YISHUN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2013

CHEMISTRY **8872/02**

HIGHER 1 **21 August 2013**

Paper 2: Structured & Free Response Questions **0800 hrs – 1000 hrs**

2 hours

Additional materials:
Writing papers, Data Booklet, graph paper

INSTRUCTIONS TO CANDIDATES

Write your name and CTG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

You are advised to show all working in calculations.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Paper 1	
Total	**/30**
Paper 2	
Section A	/40
B5	/20
B6	/20
B7	/20
Total	**/80**
Overall	**/110**

This question paper consists of **11** printed pages.

Section A

Answer **all** the questions in this section in the spaces provided.

1 Olympic torches have used a variety of fuels since the first Olympic games in 776 B.C. Propane gas was the fuel chosen for the Beijing Olympics in 2008.

(a) **(i)** Define the enthalpy change of combustion of propane.

....

....

(ii) In an experiment, 100 cm^3 of propane was burnt to raise the temperature of 250 cm^3 of water from 25 °C to 31 °C. Given that the process is 70% efficient, determine the enthalpy change of combustion of propane.

(iii) Using your answer from **(a)(ii)** and the following data, calculate the enthalpy change of formation of propane.

enthalpy change of combustion of carbon = -394 kJ mol^{-1}
enthalpy change of combustion of hydrogen = -286 kJ mol^{-1}

[6]

(b) The Beijing torch is made of an alloy of magnesium. Magnesium has several naturally occurring isotopes.

(i) Write the electronic arrangement of Mg: ....

(ii) A sample of magnesium had the following isotopic composition:
^{24}Mg, 78.60%; ^{25}Mg, 10.11%; ^{26}Mg, 11.29%

Calculate the relative atomic mass, A_r, of magnesium in the sample.

[2]

Antimony, Sb, proton number 51, is another element used in alloys.

Magnesium and antimony both react when heated separately in chlorine.

When a sample of antimony was heated in chlorine under suitable conditions, a chloride **A** was formed. It consists of 53.4% Sb and 46.6% C*l*. The relative molecular mass of the chloride is 228.5.

(c) Determine the empirical formula of chloride **A** and hence, deduce its molecular formula.

[2]

(d) The chloride **A** melts at 73.4 ^{o}C while magnesium chloride melts at 714 ^{o}C.

With reference to the bonding and structure in each of the compounds, explain the difference in their melting points.

....

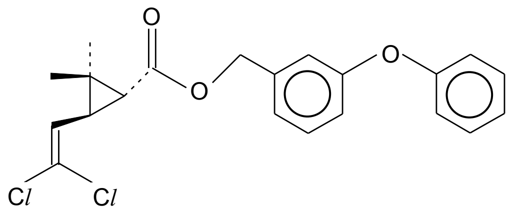
....

....

.... [2]

[Total:12]

2 Pyrethroids are insecticides which are powerful against insects but are usually harmless to mammals. The *pyrethoid permethrin* contains a mixture of stereoisomers, one of which is *biopermethrin*.



biopermethrin

(a) **(i)** Give the molecular formula of *biopermethrin*.

....

(ii) Explain why the C=C bond in *biopermethrin* does **not** give rise to geometric isomers.

....

....

....
[2]

(b) *Biopermethrin* can undergo reactions to form alcohols. One such reaction is the hydrolysis of the ester. Apart from the ester, identify the two functional groups in *biopermethrin* that will produce alcohols and state the type of reaction each functional group undergoes. In addition, give the reagents and conditions for each reaction.

Functional group: ....

Type of reaction: ....

Reagents and conditions: ....

Functional group: ....

Type of reaction: ....

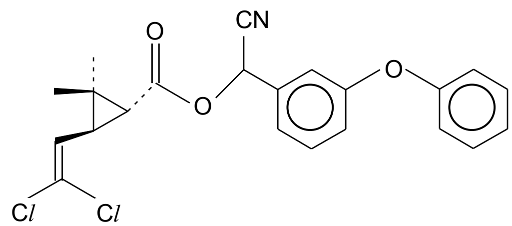
Reagents and conditions: ....
[4]

(c) Having synthesised *biopermethrin*, chemists set out to prepare even better pesticides that will be less harmful to the environment. Suggest a change in the structure of *biopermethrin* that will reduce its reactivity. Explain your answer.

....

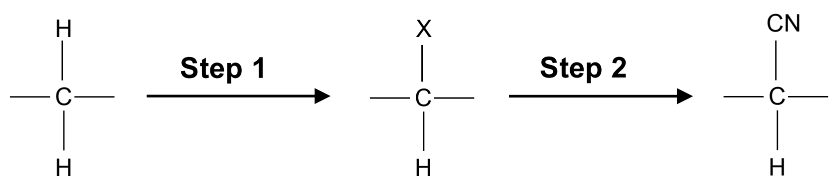
.... [1]

(d) The substance *biocypermethrin* was found to be more efficient than *biopermethrin*.



biocypermethrin

The formation of *biocypermethrin* from *biomethrin* can be shown in the following sequence.



(i) State the type of reaction for **step 1**.

....

(ii) State the reagents and conditions used for **steps 1** and **2**.

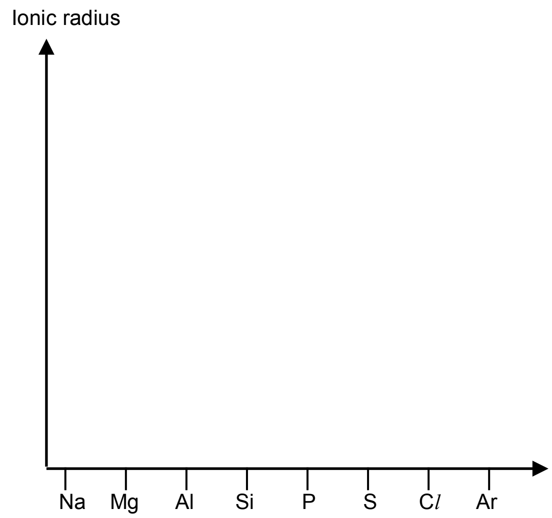
Step 1 ....

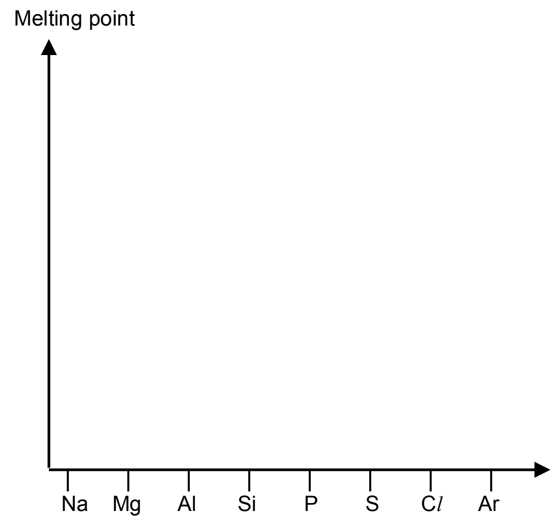
Step 2 ....

[3]

[Total:10]

3 **(a)** Complete these sketches for elements of the third period to show how each property changes along the period.





[2]

(b) Briefly explain the shapes of each of your sketches.

Ionic radius

....

....

....

.... [2]

Melting point

....

....

....

.... [2]

[Total:6]

4 Carbon dioxide (CO_2) is the primary greenhouse gas emitted through human activities and it makes up about 0.04 % of the Earth's atmosphere. Carbon dioxide is naturally present in the atmosphere as part of the Earth's carbon cycle (the natural circulation of carbon among the atmosphere, oceans, soil, plants, and animals).

In the carbon cycle, carbon dioxide is produced through animal respiration. In mammals (and many other animals), the glucose and oxygen needed for respiration are carried round the body to the cells in the bloodstream, and the bloodstream also removes the carbon dioxide and water produced by the cells' respiration. The carbon dioxide produced by respiration is a waste product and must be removed from the bloodstream, but the water produced by respiration is called metabolic water and this can be used by the body. In addition, the typical daily food requirement of a human can be considered to be the equivalent of 1.20 kg of glucose, $C_6H_{12}O_6$.

Human activities are altering the carbon cycle—both by adding more CO_2 to the atmosphere and by influencing the ability of natural sinks, like forests, to remove CO_2 from the atmosphere. While CO_2 emissions come from a variety of natural sources, human-related emissions are responsible for the increase that has occurred in the atmosphere since the industrial revolution.

The main human activity that emits CO_2 is the combustion of fossil fuels (coal, natural gas, and oil) for energy and transportation, although certain industrial processes and land-use changes also emit CO_2. When fossil fuels are burnt to release energy, carbon dioxide and water are also produced. The hydrocarbon octane, C_8H_{18}, can be used to represent the fuel burned in motor cars and the density of octane is 0.700 g cm^{-3}. A typical fuel-efficient motor car uses about 4.00 dm^3 of fuel to travel 100 km.

(a) **(i)** Construct a balanced equation for the complete oxidation of glucose.

....

(ii) Use the equation in **(a)(i)** to calculate the amount of CO_2 produced by one person in one day.

(iii) In 2012, the world population is estimated to be 6.82 x 10^9. Calculate the total mass of CO_2 produced by this number of people in one day. Give your answers in tonnes. [1 tonne = 1000 kg]

[5]

(b) **(i)** Construct a balanced equation for the complete combustion of octane.

....

(ii) Calculate the amount of octane present in 4.00 dm^3 of octane.

(iii) Calculate the mass of CO_2 produced when the fuel-efficient car is driven for a distance of 100 km.

[5]

(c) Using your answer in **(a)(iii)**, calculate how many kilometres the same fuel-efficient car would have to travel in order to produce as much CO_2 as is produced by the respiration of 6.82×10^9 people during one day.

[2]
[Total:12]

Section B

Answer **two** questions from this section on separate answer paper.

5 **(a)** Halogenoethanes react with ammonia to produce amines which are weak Bronsted bases.

$$CH_3CH_2X + NH_3 \rightarrow CH_3CH_2NH_2 + HX \qquad \text{where X = C}l\text{, Br or I}$$

(i) Arrange the halogenoethanes in increasing reactivity towards ammonia.
Explain your answer.

(ii) State the conditions required for this reaction.

(iii) Draw a dot-and-cross diagram to show the bonding in an ammonia molecule and use the VSEPR (valence shell electron pair repulsion) theory to predict its shape and bond angle. [7]

(b) A student carried out an investigation with aqueous solutions of ethylamine, $CH_3CH_2NH_2$ and hydrochloric acid.

(i) Explain what is meant by a *weak Bronsted base*.

(ii) What is the conjugate acid formed from ethylamine, $CH_3CH_2NH_2$?

(iii) The pK_b value of ethylamine, $CH_3CH_2NH_2$ is 3.36.
Calculate the pH of 0.010 mol dm^{-3} of ethylamine, $CH_3CH_2NH_2$. [6]

(c) The student prepared two solutions.

- Solution **B** was made by mixing 25 cm^3 of 0.10 mol dm^{-3} of dilute hydrochloric acid with 50 cm^3 of 0.10 mol dm^{-3} of ethylamine. Solution **B** is a buffer solution.
- Solution **C** was made by mixing 50 cm^3 of 0.10 mol dm^{-3} of dilute hydrochloric acid with 25 cm^3 of 0.10 mol dm^{-3} of ethylamine. Solution **C** is **not** a buffer solution.

(i) Explain why solution **B** is a buffer solution while solution **C** is not.

(ii) Write balanced equations to show how solution **B** is able to maintain pH when small amounts of acid and base are added to it. [5]

(d) The student measured the pH of water as 7.0 at 25 °C. She then warmed the water to 40 °C and measured the pH as 6.7.

What do these results tell you about the tendency of water to ionise as it gets warmer? Explain your reasoning in terms of LeChatelier's Principle.

. [2]

[Total:20]

6 **(a)** In order to investigate the kinetics of the hydrolysis of an ester $RCOOCH_3$, the concentration of the carboxylic acid, RCOOH produced was determined.

$$RCOOCH_3(aq) + H_2O(l) \rightarrow RCOOH(aq) + CH_3OH(aq)$$

When 0.200 mol dm^{-3} of the ester was hydrolysed in the presence of dilute sulfuric acid as catalyst, the following results were obtained.

Time / s	[RCOOH] / mol dm^{-3}
0	0
25	0.048
50	0.085
75	0.112
100	0.133
125	0.149

(i) Explain the term *order of reaction*.

(ii) By using a suitable graph, explain why the experimental results indicate that the reaction is first order with respect to the ester.

(iii) Why is it not possible to determine the order with respect to water in this experiment?

(iv) Use the following data to deduce the order of reaction with respect to hydrochloric acid.

[$RCOOCH_3$] / mol dm^{-3}	[HCl] / mol dm^{-3}	Relative rate
0.10	0.10	1
0.15	0.20	3
0.20	0.10	2

(v) Write the rate equation for the reaction and state the units of the rate constant.

(vi) With the aid of a Boltzman distribution and reference to the collision theory, explain how a rate of reaction would change when temperature is increased. [13]

(b) The above hydrolysis reaction is reversible. In an experiment, the equilibrium concentrations of the various species at 25 °C are as follows.
[$RCOOCH_3$ (aq)] = 0.188 mol dm^{-3}
[RCOOH (aq)] = 0.125 mol dm^{-3}
[CH_3OH (aq)] = 0.376 mol dm^{-3}

Solid RCOOH is added to the equilibrium mixture. When the equilibrium is re-established at 25 °C, the concentration of RCOOH (aq) becomes 0.200 mol dm^{-3}.

Calculate the concentration of CH_3OH when equilibrium is re-established at 25 °C. Assume that there is no change in the total volume of the mixture. [3]

(c) Suggest a method by which ethyl propanoate could be distinguished from propyl ethanoate by a chemical test. Give reagents and conditions for the reactions and state what you would observe. Include balanced equations for any reactions that occur. [4]

[Total:20]

7 **(a)** Using the oxides of magnesium, silicon and phosphorus as examples, describe the reactions of the oxides of elements in the third period of the Periodic Table with water.
Include balanced equations for any reactions. [4]

(b) Sodium nitrite, $NaNO_2$ is used as a preservative in meat products such as sausages and ham. In an acidic solution, the nitrite ion is converted to nitrous acid, HNO_2, which reacts with manganate(VII) ion to form nitrate ion, NO_3^-.

(i) Write a half equation for the reaction of the nitrate ion, NO_3^-.

(ii) Give the oxidation states of N in this reaction and use them to explain why NO_3^- is acting as a reducing agent.

(iii) Give a balanced equation for the redox reaction between nitrite ions and manganate(VII) ions in acid solution.

(iv) A 1.00 g sample of a water-soluble solid containing sodium nitrite was dissolved in dilute hydrochloric acid and titrated with 0.0100 mol dm^{-3} $KMnO_4$ solution. The titration required 12.5 cm^3 of $KMnO_4$ solution.

Calculate the percentage by mass of sodium nitrite in the sample. [6]

(c) Compound **P**, C_9H_8O is the main component of cinnamon oil.

An orange precipitate is observed when **P** is warmed with 2,4-dinitrophenylhydrazine. In addition, **P** gives a red precipitate with Fehling's solution and decolourises aqueous bromine.

When **P** was refluxed with acidified potassium manganate(VI), compound **Q**, $C_7H_6O_2$ and carbon dioxide are formed. **Q** is soluble in both aqueous sodium hydroxide and sodium carbonate.

Treatment of **P** with sodium boronhydride produces compound **R** which gives hydrogen gas with sodium metal. When **R** reacts with hydrogen chloride, compound **S** is formed.

Identify and suggest the structures for **P**, **Q**, **R** and **S**. Show how you deduced these structures by explaining the reactions that are occurring. [10]

[Total:20]

~ END OF PAPER ~